

■ SolveAlways

`SolveAlways[eqns, vars]` gives the values of parameters that make the equations *eqns* valid for all values of the variables *vars*.

Equations are given in the form *lhs* == *rhs*. ■ Simultaneous equations can be combined either in a list or with `&&`. ■ A single variable or a list of variables can be specified. ■ Example:

`SolveAlways[a x + b == 0, x] → {{b -> 0, a -> 0}}`. ■ `SolveAlways` works primarily with linear and polynomial equations. ■ `SolveAlways` produces relations between parameters that appear in *eqns*, but are not in the list of variables *vars*. ■ `SolveAlways[eqns, vars]` is equivalent to `Solve[!Eliminate[!eqns, vars]]`. ■ See page 408.

■ See also: `Eliminate`, `Solve`, `Reduce`, `AlgebraicRules`.